

Activity - 1
Section - 3J
Date: 21-Aug-25

Student Roll No. _____

Student Name: _____

29

Q:1 - If set $A = \{1,2,3\}$ and set $B = \{4,5,6\}$ then write domain and co-domain for the below functions. Also mention if a function is valid or not. $(2+1) \times 3$

a) $f(x) = \{(1,4), (2,4), (3,5)\}$

b) $f(x) = \{(1,4), (1,5), (3,6)\}$

c) $f(x) = \{(1,6), (2,4), (3,5)\}$

(a) Domain = $\{1, 2, 3\}$, Co-domain = $\{4, 4, 5\}$ (Valid function)

(b) Not a valid function

(c) Valid function, Domain = $\{1, 2, 3\}$, Co-domain = ~~$\{6, 4, 5\}$~~
 $\{4, 5, 6\}$



Q:2 - Develop languages based on given rules where $\Sigma = \{a, b\}$

a) L_1 = All strings ending at 'ab' where $m \leq 3$

b) L_2 = All strings that starts with an 'a' where $m \leq 4$

c) L_3 = Palindrome strings with $m \leq 3$

$L_1 = \{ab, aab, bab\}$ ~~$aaab, abab, baab, bbab$~~

$L_2 = \{a, aa, ab, aaa, aab, aba, abb, aaaa, aaab, aaba, aabb, abaa, abab, abba, a$

$L_3 = \{\Lambda, a, b, aa, bb, aaa, aba, bab, bbb\}$